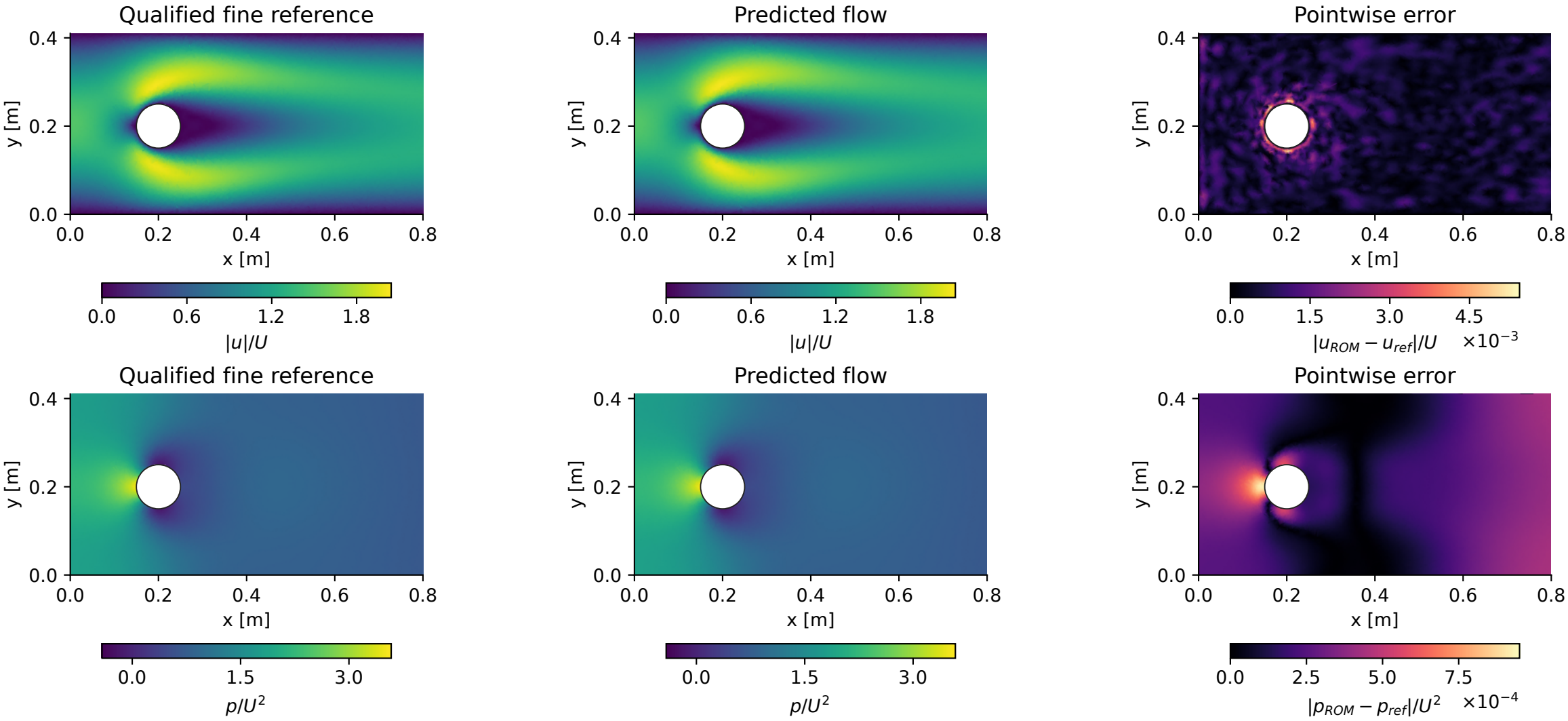

Surface-traction-trained neural-bank ROM: flow-solve output — Re = 16.875
Velocity (top) and pressure (bottom); near-cylinder view
Qualified fine reference
Predicted flow
Pointwise error
y [m]
x [m]
0.0
0.2
0.4
0.6
0.8
0.0
0.6
1.2
1.8
|u|/U
0.0
1.5
3.0
4.5
|u_ROM − u_ref|/U
×10^−3
Qualified fine reference
Predicted flow
Pointwise error
0.0
1.5
3.0
p/U²
0.0
2.5
5.0
7.5
|p_ROM − p_ref|/U²
×10^−4
Four neural velocity modes plus FE pressure/stability spaces. Actual reduced solve; consult full-rule metrics for accuracy and conservation.
Display interpolation; error scales differ. Errors against stored reference do not certify equal continuum accuracy. Development cases.